

ANL559 Week 1 Lab Exercise And Case Study

Updated handout generated from the current Markdown lab exercise and Week 1 case pack.

Week 1 Lab Exercise - Reporting Fire Drill

Purpose

Students learn that an AI-generated report is not trustworthy until the metric definitions, source files, checks, and rejected agent assumptions are visible.

Use this sheet with the Week 1 section of `Weekly_Case_Packs.md`. The case pack provides the fuller stakeholder story, intentional traps, and solution guidance.

Materials

- Starter_Kit/sample_data/week1_customers.csv
- Starter_Kit/sample_data/week1_sales.csv
- Starter_Kit/sample_data/week1_support.csv
- Starter_Kit/sample_data/week1_metric_definitions.md
- Starter_Kit/templates/work_order_template.md
- Starter_Kit/templates/ai_use_log_template.md
- Starter_Kit/templates/defence_note_template.md

Case Setup For Students

Northstar Retail's COO needs the weekly KPI pack by 10:00. The usual analyst is away, and last week's pack was created manually from CSV files. Your team must create a repeatable workflow that another analyst could rerun.

The main risk is the phrase "active customer." Product often means recent login. The COO means paid account with active subscription status. If the agent chooses the wrong definition, the report can look polished and still be wrong.

By the end of class, each team should be able to show:

- A work order that controlled the agent's task.
- The metric definitions used by the workflow.
- KPI output checked against known sample totals.
- One rejected or corrected agent assumption.
- A short defence of what can be trusted.

Learning Outcomes

After this exercise, students should be able to:

- Convert a vague reporting request into an agent work order.
- Treat metric definitions as acceptance criteria.
- Design smoke tests that catch plausible agent errors.
- Explain why one agent output was accepted, rejected, or modified.

Studio Flow

Time	Activity	Concrete output
0-10	Case briefing and file orientation	Students identify stakeholder, deadline, files, and risk phrase.
10-25	Exercise 1	Metric contract table.
25-40	Exercise 2	Revised agent work order.
40-65	Exercise 3	Smoke checks and first KPI output.
65-80	Exercise 4	AI-use-log entry.
80-90	Exercise 5	60-second defence.

If running the full three-hour seminar, use this sheet as the guided core inside the longer Build / Break / Defend schedule from `Complete_Instructor_Guide.md`.

Exercise 1 - Metric Contract Clinic

Time: 15 minutes

Student task: In pairs, list the metrics the COO probably expects in the Monday KPI pack. For each metric, write the definition, source file, formula, and one possible ambiguity.

Suggested table:

Metric	Approved definition	Source	Formula	Possible ambiguity
Paid active accounts				
Net June revenue				
Open support tickets				

Instructor prompt: "If the agent used a different definition, how would you know before the COO sees the report?"

Expected evidence: A metric-definition table with at least one ambiguity clearly named.

Look for: Students should name who owns the definition, not only the formula. For example, paid active accounts is a COO or Finance definition, while recent login is a Product diagnostic.

Exercise 2 - Rewrite the Weak Prompt

Time: 15 minutes

Weak prompt:

Make me a KPI report from these CSVs.

Student task: Rewrite the prompt as an agent work order. It must include goal, inputs, acceptance criteria, and constraints.

Minimum requirements:

- Name all source files.
- State that the agent must not invent metric definitions.
- Require at least three smoke tests.
- Require a short explanation of any assumptions.

Instructor note: Push students to specify output format. A vague "make a report" prompt often produces a polished but uncheckable artefact.

Acceptance check: The revised prompt should be specific enough that a peer can predict the file names, metrics, and tests before the agent answers.

Exercise 3 - Smoke-Test Hunt

Time: 25 minutes

Student task: Use the sample files to create at least three checks that would catch obvious errors.

Known checks:

- Paid active accounts should be 6.
- Recent-login customers should be 7.
- Gross June revenue should be 2860.
- Net June revenue should be 2660.
- Open support tickets should be 3.

Challenge: Write one check that would fail if the agent used recent-login customers instead of paid active accounts.

Expected evidence: A script, notebook, spreadsheet, or plain-language check list showing input, formula, expected value, and actual value.

Instructor checkpoint: Ask teams to explain the defect each smoke test is designed to catch. A check without a defect story is usually too generic.

Exercise 4 - Agent Failure Log

Time: 15 minutes

Student task: Ask the agent to draft the KPI workflow. Identify one suggestion you would reject or modify.

Log it using:

Agent suggestion	Risk	Evidence used	Team decision

Instructor prompt: "The best log entry is not the longest. It is the entry that shows judgement."

Good entries name a concrete agent contribution:

- Accepted: file-loading structure, grouping logic, output format.
- Rejected: invented active-customer definition, gross/net confusion, unsupported executive summary.
- Modified: test names, assumption notes, error handling.

Exercise 5 - Defence Rehearsal

Time: 10 minutes

Each pair gives a 60-second defence:

1. What number do you trust?
2. What did the agent get wrong or almost get wrong?
3. Which check would catch the problem next time?

Listeners ask one question: "What would still make this report unsafe?"

End-Of-Class Bundle

Teams should submit or save:

- `work_order.md`
- Metric-definition table.
- KPI workflow artefact: script, notebook, spreadsheet, or documented manual workflow.
- Smoke-test evidence with expected and actual values.
- One AI-use-log entry.
- Defence note beginning: "I trust this KPI pack because..."

Fallback Routes

If students cannot run code:

- Use spreadsheet formulas for the three CSV files.
- Write smoke tests as a validation table.
- Ask the agent to explain the calculation logic, then audit the explanation against the metric definitions.

If students finish early:

- Add a data-freshness check.
- Add a stop condition if any metric check fails.
- Generate a COO summary that cites only checked values.

Instructor Solution Notes

Strong teams usually separate paid active accounts from recent login customers. If a team reports only one "active" number, ask who owns the definition and whether a logged-in cancelled customer should count.

Common mistakes:

- Treating the sample metric definitions as optional reading.
- Reporting gross revenue when the stakeholder expects net revenue.
- Writing tests after seeing the answer without explaining what defect the test prevents.
- Logging "AI helped with code" instead of a specific accepted or rejected output.

Extension Options

- Add a generated executive summary that cites only checked numbers.
- Create a scheduled-run checklist.
- Add a "data freshness" check: no source file should be older than the reporting date.

Week 1 Case Study - Reporting Fire Drill

Case Snapshot

Item	Detail
Stakeholder	COO
Business pressure	The Monday KPI pack is due by 10:00 and the usual analyst is unavailable.
Student artefact	Repeatable KPI workflow with metric definitions and smoke tests.
Files	week1_customers.csv, week1_sales.csv, week1_support.csv, week1_metric_definitions.md
Main agent trap	The agent treats "active customer" as recent login instead of paid active account.
Defence evidence	Work order, metric contract, checked KPI output, smoke-test result, AI-use log.

Student Brief

Northstar Retail sends a weekly KPI pack to the COO. The pack is normally assembled through manual copy-paste from customer, sales, and support files. This week the analyst who owns the process is on leave, and the COO still expects the pack by 10:00.

Your team must use an AI agent to create a repeatable workflow from the starter CSV files. The COO does not need a beautiful dashboard yet. The COO needs numbers that can be rerun, checked, and defended.

The risky phrase is "active customer." Product often uses recent login. The COO means a paid account with active subscription status. If your workflow uses the wrong denominator, the report is not safe to send.

Approved Business Definitions

Metric	Approved definition	Expected check in sample data
Paid active accounts	subscription_status = active and is_paid = true	6
Recent-login customers	last_login_date within 30 days of 2026-06-26	7
Gross June revenue	Sum gross_amount for June 2026 orders	2860
Net June revenue	Sum gross_amount - refund_amount for June 2026 orders	2660
Open support tickets	Blank closed_date	3

The paid-active and recent-login populations differ. For example, a cancelled or past-due customer can still log in, while a paid active customer may not have logged in recently. A reliable workflow should make this difference visible.

Instructor Setup

Before class:

- Put the four Week 1 files in a shared folder or repo.
- Prepare a weak prompt on a slide: "Make me a KPI report from these CSVs."
- Prepare one flawed agent output or live-demo prompt where the agent defines active customers from last_login_date.

- Open Starter_Kit/templates/work_order_template.md and Starter_Kit/templates/metric_definition_template.md.

Failure cards to release after Sprint A:

- "The COO asks why active customers are 7 when Finance expected 6."
- "A refund was posted in June but the report shows gross revenue."
- "Support asks whether open tickets are based on opened date or unresolved status."

Studio Path

Phase	Student action	Instructor coaching
Build	Write an agent work order naming files, definitions, output format, and acceptance checks.	Ask where the metric owner appears in the work order.
Build	Ask the agent to draft a script, notebook, spreadsheet, or report workflow.	Encourage a rough first version; polish is not the objective.
Break	Compare the agent's metric definitions with week1_metric_definitions.md.	Ask which definition the agent invented or assumed.
Break	Run smoke checks for paid active accounts, revenue, and open tickets.	Ask students to make at least one check fail before they fix it.
Defend	Write a short note explaining which KPI can be trusted and which assumption was rejected.	Push for evidence-linked language, not "the script works."

Expected Evidence

Minimum evidence:

- Agent work order that names the files and approved definitions.
- Metric-definition table with at least three metrics.
- KPI output with the known checks above.
- One AI-use-log entry naming an accepted and rejected agent contribution.
- Defence note: "I trust this KPI pack because..."

Strong evidence:

- A check that would fail if recent login is used as paid active account.
- A notes section listing assumption, owner, and unresolved risk.
- A rerun instruction someone else can follow.

Solution Guidance

Do not focus on a single implementation style. A spreadsheet, notebook, or Python script can all satisfy the case if the definitions and checks are visible.

Coach toward these insights:

- C103 and C106 are recent-login or recently active in product behaviour but are not paid active accounts.
- Gross and net revenue are both valid metrics, but they answer different questions.
- A smoke test should be tied to a known defect. "Paid active accounts should equal 6" is stronger than "the script ran."
- The agent can generate useful code structure, but it cannot own the metric definition.

Low-code route:

- Students compute counts and sums in a spreadsheet.

- They write smoke tests as formulas or a manual validation table.

Stretch route:

- Add a freshness check for each source file.
- Generate a one-page COO summary that cites only checked numbers.
- Add a "stop the send" condition if any check fails.